Amagi HARA v1.1 — Hazard Analysis and Risk Assessment

VERSION 1.1 — OCTOBER 2026

> Systematic hazard identification and risk assessment for the AMAGI Architectural Class per IEC 61508-1/-5 (risk graph method, Annex D). Covers hazard identification, risk classification, SIL assignment, and hazard-to-invariant traceability for all identified safety-relevant scenarios. Aligned with: Framework v2.0, NIPU v1.1, MEM v1.1, Master Parameter Sheet v2.1, FMEDA v1.1, Traceability Matrix v1.1, Master Glossary v1.0, S-APL v2.0, Charter v1.2.

Alexey Mikhailovich Burlai AGIAM Technologies Ltd (In Formation) ORCID: 0009-0001-4679-5967
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1. Executive Summary

This document presents the first formal Hazard Analysis and Risk Assessment (HARA) for the AMAGI Architectural Class, establishing the hazard-to-invariant traceability required by IEC 61508 Part 1 §§7.4–7.6. The AMAGI architectural class is designed to provide hardware-enforced AI safety in autonomous cyber-physical systems (CPS), targeting a configuration-dependent safety-integrity capability — SIL 2 on the 1oo1 reference configuration (target PFDavg 6.9×10^{-5}) and SIL 3 on the enhanced 1oo2 configuration (target PFDavg 6.9×10^{-6}, Phase 2) — per the integrity capability formula (FMEDA v1.1 §8; MPS v2.1 §3.1). This HARA identifies eight distinct hazard scenarios that are reasonably foreseeable in the life-critical domains where AMAGI-class systems are intended to be deployed, including medical implants, autonomous AI agents, nuclear control systems, and avionics platforms.

The eight hazard scenarios are systematically mapped to the six AMAGI hardware-enforced invariants (INV-001 through INV-006), demonstrating that each invariant provides risk reduction for at least two independent hazard categories. SIL targets are derived using the risk graph method prescribed by IEC 61508-5 Annex D, with conservative assumptions reflecting the worst-case threat model in which the AI model is treated as a hostile insider capable of provoking any hazard at will. This conservative posture ensures that the AMAGI architecture provides maximum risk reduction regardless of the AI's behavioral state, thereby satisfying the defense-in-depth principle required by IEC 61508-1 §7.6.

The analysis confirms that four of the eight identified hazards require SIL 3 integrity (H-01, H-02, H-05, H-06); two are assessed in the range SIL 2–3 and are conservatively treated as SIL 3 for deployment-configuration selection (H-03, H-07); the remaining two require SIL 2 (H-04, H-08). INV-001 (Physical Non-Bypassability) emerges as the most critical invariant, providing primary mitigation for six of the eight hazards. INV-005 (Complete Mediation) counters five hazards, and INV-004 (Controlled Mutability of Critical Policies) counters four. These coverage ratios justify the non-waivable status of INV-001, INV-005, and INV-006 within the AMAGI architectural class definition, as their removal would leave one or more SIL 3 hazards without adequate hardware-enforced mitigation.

2. Scope and Purpose

This HARA covers the AMAGI Target of Evaluation (TOE), which encompasses the following hardware components: the Safety Control Core (SCC), the Memory Policy Enforcer (MPE), the **Memory Access Graph (MAG)** stored in eFuse/OTP, the **Access Trigger Layer (ATL)** — implementing the TTP delay at the memory level (0.5 ns–1 µs) together with the audit transaction

logger (Audit Diode) — the Self-Healing Memory Fabric (SHMF), the Amagi-Q post-quantum cryptographic engine, the Executive Gateway (EGW), and the Core Root of Trust for Measurement (CRTM). The TOE operates in life-critical domains including medical implants (e.g., drug delivery pumps, cardiac rhythm management devices), autonomous AI agents with real-world authority (e.g., robotic surgery, autonomous vehicles), nuclear control systems (e.g., reactor trip systems, coolant flow regulation), and avionics platforms (e.g., flight control actuation). Each of these domains imposes unique safety requirements that this HARA addresses through a domain-independent hazard identification and risk assessment methodology.

The hazard analysis follows the structured process defined by IEC 61508-1 §7.4 (Hazard Identification), §7.5 (Risk Assessment), and §7.6 (SIL Assignment). The purpose of this document is fourfold: (a) to identify all reasonably foreseeable hazards arising from the operation of AMAGI-class systems in their intended deployment domains; (b) to assess the risk associated with each hazard using severity, exposure, and avoidability parameters as defined by the risk graph method in IEC 61508-5 Annex D; (c) to assign SIL targets to each hazard based on the assessed risk; and (d) to trace each hazard to one or more AMAGI invariants that mitigate it, thereby establishing the hazard-to-invariant traceability required for functional safety certification.

The scope of this HARA is limited to hazards arising from the normal operation and reasonably foreseeable misuse of the AMAGI TOE. Hazards from beyond the certification boundary (e.g., physical destruction of the chip, electromagnetic pulse attacks beyond the specified environmental envelope) are explicitly excluded and documented as residual risks in Section 9. The HARA does not address software-level hazards within the cognitive clusters, as these are contained by the hardware-enforced isolation boundary and cannot directly affect actuator paths without passing through the SCC-gated security perimeter.

Terminology note (v1.1): «Memory Access Graph (MAG)» and «Access Trigger Layer (ATL)» are the canonical expansions per Master Glossary v1.0 (CCR-01); the v1.0 readings «Mutable Action Gate» and «Audit Trail Log» are withdrawn. The audit function (Audit Diode) is embedded in the TTP Engine/ATL per MEM v1.1 §3.3; the H-04 traceability (SR-07/SR-08 → INV-006) is unchanged.

Methodology note (v1.1): risk classification uses the risk graph method per IEC 61508-5 Annex D. ISO 26262-3/ASIL terminology is a domain deployment mapping and is out of scope of this document; see MPS v2.1 §3.4.

3. Methodology

The risk assessment methodology employed in this HARA follows the risk graph method as prescribed by IEC 61508-5 Annex D. The risk graph uses four parameters to determine the required Safety Integrity Level for each identified hazard. These parameters are: Consequence (C), which characterizes the severity of the potential harm resulting from the hazard; Frequency/Exposure (F), which characterizes how often the hazardous situation is expected to occur; Possibility of Avoidance (P), which characterizes the ability of the operator or affected party to avoid the harm once the hazardous event has occurred; and Probability of Unwanted Occurrence (W), which characterizes the likelihood that the hazard will manifest without the safety function in place.

The consequence parameter C is rated on a four-point scale: C1 (minor injury), C2 (serious permanent injury to one or more persons, or death of one person), C3 (death of several persons), and C4 (catastrophic: multiple fatalities). For AMAGI deployments in medical, nuclear, and avionics domains, the worst-case consequence is C4 because uncontrolled actuator activation can result in multiple simultaneous fatalities (e.g., a nuclear reactor excursion, an aircraft loss of control, or a systemic failure of a medical device fleet). The frequency parameter F distinguishes between F1 (rare exposure, less than once per year) and F2 (frequent to permanent exposure). For AMAGI systems under continuous AI control, F2 is the appropriate rating because the AI model has persistent access to the actuator pathway. The avoidance parameter P distinguishes between P1 (possible to avoid) and P2 (almost impossible to avoid). For autonomous AI systems operating

without a human-in-the-loop, P2 is the appropriate rating because the speed and opacity of AI decision-making make it practically impossible for a human operator to intervene before harm occurs. The probability parameter W is rated on a three-point scale: W1 (very low), W2 (low), and W3 (relatively high). For the AMAGI threat model, which assumes the AI model can behave as a hostile insider, W is always set to W3 (high probability of unwanted occurrence). This conservative approach ensures that the architecture provides maximum risk reduction regardless of the AI's behavioral state, and it reflects the fundamental design assumption that software cannot be trusted to enforce safety constraints that only hardware can guarantee.

Conservative Threat Model Justification: The assignment of W3 to all hazard scenarios is a deliberate conservative choice mandated by the AMAGI architectural philosophy. In conventional functional safety analysis, W is calibrated based on historical failure data for the specific component under analysis. However, for AI-driven systems, no statistically significant historical failure data exists, and the behavior space of a large language model or neural network is effectively unbounded. By assuming W3, the HARA ensures that the SIL targets are not dependent on assumptions about AI reliability, which remains an open research question. This approach is consistent with IEC 61508-1 §7.4.2.2, which requires that the hazard analysis consider "all reasonably foreseeable hazards," including those arising from unexpected software behavior.

4. System Boundary and Operating Modes

The AMAGI TOE lifecycle encompasses four distinct phases, each with its own hazard profile and risk assessment considerations. The Fabrication phase covers the manufacturing and initial programming of the AMAGI chip, including the burning of eFuse/OTP bits for the MAG policy matrix, the verification of the SCC firmware hash against the CRTM golden hash, and the physical sealing of the chip package. Hazards during fabrication primarily concern supply chain integrity (e.g., hardware Trojan insertion, unauthorized modification of MAG bits) and are mitigated by post-silicon validation procedures documented in Section 8.

The Deployment phase covers the installation and commissioning of the AMAGI system in its target environment. This phase includes the physical integration of the chip into the host system, the activation of the physical presence switch for Tier 2 OTA policy updates, and the initial boot-time integrity verification. The Normal Operation phase is the primary operational state in which the AMAGI system executes its intended functions under the governance of the SCC policy engine. Normal Operation is further subdivided into four operating sub-modes: Active (cognitive clusters processing, SCC mediating actuator-bound commands), Quiescent (idle state, clusters in sleep mode awaiting external triggers), Emergency (cutoff triggered by SCC, actuator path physically disconnected, AFC activated if present), and Fallback (AFC active, maintaining minimal-safe operational state independently of SCC).

The Maintenance phase covers Tier 2 policy updates via Physical-Presence-Verified OTA, which require the activation of a physical presence switch on the board. In Tier 1 (Hard Assurance) deployments, the Maintenance phase is physically inaccessible because the physical presence switch is permanently sealed. The environmental conditions for Normal Operation are specified as: operating temperature range −40°C to +125°C (industrial grade), supply voltage range 0.9V to 1.2V, and electromagnetic immunity per IEC 61000-4 Level 4. These conditions define the environmental envelope within which the AMAGI invariants are guaranteed to hold; hazards arising from operation outside this envelope are documented as beyond-certification-boundary risks in Section 9.

Operating Mode	Description	Hazard Relevance
Active	Cognitive clusters processing; SCC mediating all actuator-bound commands	H-01, H-03, H-04, H-05
Quiescent	Idle state; clusters in sleep mode awaiting external triggers	Low risk — no actuator path active

Operating Mode	Description	Hazard Relevance
Emergency	SCC cutoff triggered; actuator path physically disconnected	H-06 (kill switch failure)
Fallback	AFC active; minimal-safe state maintained independently of SCC	Residual risk only — AFC override

Table 1: AMAGI operating sub-modes and hazard relevance

Phase	Activities	Applicable Hazards
Fabrication	Chip manufacturing, OTP/MAG programming, CRTM golden hash verification, package sealing	H-02, H-07
Deployment	Physical integration, boot-time integrity check, initial policy activation	H-02, H-07
Normal Operation	Active/Quiescent/Emergency/Fallback sub-modes operating under SCC governance	H-01, H-03, H-04, H-05, H-06
Maintenance	Tier 2 OTA policy updates via physical presence switch; Tier 1: no maintenance access	H-08

Table 2: AMAGI lifecycle phases and hazard applicability

5. Hazard Identification

This section identifies eight hazard scenarios that are reasonably foreseeable for the AMAGI TOE operating in its intended deployment domains. Each hazard is assigned a unique identifier (H-01 through H-08), a descriptive name, a detailed description of the hazard mechanism, the lifecycle phase(s) in which the hazard may manifest, and a consequence class per IEC 61508-5 Annex D. The hazard identification process follows the systematic approach prescribed by IEC 61508-1 §7.4, which requires consideration of all reasonably foreseeable hazards including those arising from normal operation, foreseeable misuse, and reasonably foreseeable external events. The eight hazards identified below represent the complete set of hazard scenarios within the AMAGI certification boundary; hazards outside this boundary are documented as residual risks in Section 9.

ID	Hazard	Description	Phase	Consequence
H-01	Uncommanded Actuator Activation	AI model bypasses TTP chain to directly activate physical actuator, causing harm in life-critical domain (e.g., drug overdose, reactor excursion)	Normal Operation	C4 (catastrophic: multiple fatalities)
H-02	Policy Store Corruption	MAG eFuse/OTP bits flipped during fabrication or by physical tampering, creating unauthorized access paths not sanctioned by the policy matrix	Fabrication / Normal Operation	C3–C4 (critical to catastrophic)
H-03	Bypass of Memory Access Control	Cognitive cluster accesses Secure World memory without MPE mediation, exfiltrating cryptographic keys or audit data from protected regions	Normal Operation	C3 (critical: single fatality or permanent disability)
H-04	Audit Trail Compromise	Append-only audit log overwritten or tampered with, preventing forensic analysis after a safety incident and concealing evidence of security violations	Normal Operation / Post-Incident	C2–C3 (significant to critical)
H-05	TTP Delay Subversion	Programmable delay shortened or eliminated by hardware fault or PVT drift, removing the safety time window that permits human or system intervention	Normal Operation	C4 (catastrophic)

ID	Hazard	Description	Phase	Consequence
H-06	Kill Switch Failure	Termination latch fails to activate when SCC commands cutoff, leaving the actuator path energized and the system in an unsafe state	Emergency	C4 (catastrophic)
H-07	Supply Chain Hardware Trojan	Malicious modification of MAG or SCC during fabrication, creating hidden bypass path that is undetectable by software-level integrity checks	Fabrication	C3–C4 (critical to catastrophic)
H-08	Rollback Attack on Policy	Older, vulnerable MAG policy version installed via OTA, bypassing security improvements and re-enabling previously closed attack vectors	Maintenance (Tier 2)	C3 (critical)

Table 3: AMAGI hazard identification register

The eight hazards identified above span all four lifecycle phases and cover both accidental and malicious threat origins. Hazards H-01, H-03, H-05, and H-06 are primarily operational hazards that manifest during Normal Operation or Emergency modes, while H-02 and H-07 are primarily supply chain hazards that originate during Fabrication but may not become apparent until Normal Operation. H-04 is a post-incident hazard that affects forensic analysis capability, and H-08 is a maintenance-phase hazard specific to Tier 2 deployments. The consequence classes range from C2 (significant injury) to C4 (catastrophic: multiple fatalities), reflecting the diversity of AMAGI deployment domains and the severity of harm that could result from each hazard scenario.

6. Risk Assessment and SIL Assignment

This section presents the risk assessment and SIL assignment for each of the eight identified hazard scenarios, using the risk graph method per IEC 61508-5 Annex D. Each hazard is characterized by its Consequence (C), Frequency/Exposure (F), Possibility of Avoidance (P), and Probability of Unwanted Occurrence (W) parameters. The combination of these four parameters determines the required SIL class. As discussed in Section 3, the AMAGI threat model mandates that W is always set to W3 (high probability), reflecting the conservative assumption that the AI model can behave as a hostile insider. The risk reduction factor indicates the order of magnitude by which the safety function must reduce the probability of the hazardous event.

Table 4 — Risk assessment and SIL assignment (v1.1)

Hazard	C	F	P	W	SIL target	PFDavg band (IEC 61508-1 Table 3)	Required deployment configuration
H-01	C4	F2	P2	W3	SIL 3	10^{-4} to 10^{-3}	**1oo2**
H-02	C4	F1	P2	W3	SIL 3	10^{-4} to 10^{-3}	**1oo2**
H-03	C3	F2	P2	W3	SIL 2–3 (cons. SIL 3)	10^{-4} to 10^{-3}	**1oo2** (1oo1 acceptable with a documented SIL 2 boundary justification)
H-04	C3	F2	P1	W3	SIL 2	10^{-2} to 10^{-3}	1oo1
H-05	C4	F1	P2	W3	SIL 3	10^{-4} to 10^{-3}	**1oo2**
H-06	C4	F1	P2	W3	SIL 3	10^{-4} to 10^{-3}	**1oo2**
H-07	C3	F1	P2	W3	SIL 2–3 (cons. SIL 3)	10^{-4} to 10^{-3}	**1oo2** (1oo1 acceptable with a documented SIL 2 boundary justification)
H-08	C3	F1	P1	W3	SIL 2	10^{-2} to 10^{-3}	1oo1

> Note on the PFDavg column: the v1.0 values in this column were correct as PFDavg bands, but the column was headed «Risk Reduction Factor» (corrected, H2). RRF equivalents: SIL 3 → $10^{3}..10^{4}$; SIL 2 → $10^{2}..10^{3}$. The H-03/H-07 bands correspond to the conservative SIL 3 reading (the v1.0

> values «10−3 to 10−4» denote the same SIL 3 band; listed low→high without changing the numbers).

Table 4: Risk assessment and SIL assignment per IEC 61508-5 Annex D

Detailed SIL Justification: The SIL assignments above are derived from the risk graph lookup tables in IEC 61508-5 Annex D, applied with the following rationale for each parameter.

H-01 (Uncommanded Actuator Activation): C4 because uncontrolled actuator activation in medical or nuclear domains can cause multiple simultaneous fatalities. F2 because the AI model has continuous access to the actuator pathway during Active mode, making the hazardous situation frequently present. P2 because in autonomous systems there is no human operator who can reliably intervene in the milliseconds between AI command issuance and actuator response. W3 per the AMAGI threat model. The combination C4-F2-P2-W3 yields SIL 3, requiring a risk reduction factor of 10−3 to 10−4.

H-02 (Policy Store Corruption): C4 because corrupted MAG policy could enable unrestricted actuator access. F1 because fabrication-phase corruption is a one-time event rather than a continuous exposure. P2 because corrupted policy may not be detectable until a safety incident occurs, at which point avoidance is impossible. W3 per the threat model. The combination C4-F1-P2-W3 yields SIL 3.

H-03 (Bypass of Memory Access Control): C3 because key exfiltration from Secure World could enable a single fatality through targeted attack. F2 because memory access is continuous during Active mode. P2 because the attack is silent and undetectable in real-time. W3 per the threat model. The combination C3-F2-P2-W3 yields SIL 2–3; the AMAGI architecture targets SIL 3 for this hazard as a conservative choice.

H-04 (Audit Trail Compromise): C3 because loss of forensic evidence impedes post-incident investigation and could delay identification of a safety-relevant defect. C2–C3 is the full range; C3 is used for the SIL determination. F2 because the audit trail is continuously written during Active mode. P1 because forensic analysis is typically conducted post-incident when there is time to detect and respond to tampering evidence. W3 per the threat model. The combination C3-F2-P1-W3 yields SIL 2.

H-05 (TTP Delay Subversion): C4 because removal of the safety time window could allow immediate, uncommanded actuator activation with catastrophic consequences. F1 because hardware fault or PVT drift is a rare event. P2 because the subversion is undetectable by the operator until the hazard manifests. W3 per the threat model. The combination C4-F1-P2-W3 yields SIL 3.

H-06 (Kill Switch Failure): C4 because failure to disconnect the actuator path during an emergency leaves the system in an actively dangerous state with potential for multiple fatalities. F1 because the termination latch is a high-reliability hardware component with very low failure probability. P2 because the failure mode eliminates the last line of defense, making avoidance impossible. W3 per the threat model. The combination C4-F1-P2-W3 yields SIL 3.

H-07 (Supply Chain Hardware Trojan): C3–C4 because a hardware Trojan in MAG or SCC could create a persistent, undetectable bypass path. C3 is used for SIL determination as a conservative midpoint. F1 because supply chain attacks are rare but increasingly documented. P2 because hardware Trojans are designed to be undetectable by standard software integrity checks. W3 per the threat model. The combination C3-F1-P2-W3 yields SIL 2–3; AMAGI targets SIL 3 conservatively.

H-08 (Rollback Attack on Policy): C3 because a downgraded policy re-enables previously closed attack vectors, potentially leading to a single fatality through targeted exploitation. F1 because rollback attacks require physical access to the OTA mechanism. P1 because the monotonic version counter provides a detectable indicator of rollback, enabling avoidance through verification. W3 per the threat model. The combination C3-F1-P1-W3 yields SIL 2.

7. Hazard-to-Invariant Traceability

This section presents the critical traceability matrix linking each identified hazard to the AMAGI invariants that provide mitigation. The traceability matrix is the central artifact of this HARA, as it demonstrates that no hazard is left without at least one hardware-enforced mitigation path and that the non-waivable status of certain invariants is justified by their coverage of multiple high-severity hazards. In the table below, ● indicates primary mitigation (the invariant directly prevents or contains the hazard), and ○ indicates no direct mitigation (the invariant does not address the hazard mechanism).

Hazard	INV-001	INV-002	INV-003	INV-004	INV-005	INV-006
H-01	●	○	●	○	●	○
H-02	●	○	○	●	●	○
H-03	●	○	●	○	●	○
H-04	●	○	○	○	○	●
H-05	●	●	○	○	●	○
H-06	●	○	○	○	●	○
H-07	○	○	○	●	●	○
H-08	○	○	○	●	○	●

Table 5: Hazard-to-invariant traceability matrix (● = primary mitigation, ○ = no direct mitigation)

Invariant Coverage Analysis: INV-001 (Physical Non-Bypassability) provides primary mitigation for 6 of 8 hazards (H-01, H-02, H-03, H-04, H-05, H-06), making it the single most critical invariant in the AMAGI architectural class. Its coverage spans all operational hazards that involve bypassing the hardware security boundary, confirming that the physical non-bypassability of the SCC, EGW, and CRTM is the foundational safety property upon which all other mitigations depend. INV-005 (Complete Mediation) counters 5 of 8 hazards (H-01, H-02, H-03, H-05, H-06), providing defense-in-depth by ensuring that every actuator-bound command passes through the SCC policy engine. INV-004 (Controlled Mutability of Critical Policies) counters 4 of 8 hazards (H-02, H-07, H-08, and indirectly H-01), ensuring that the policy matrix cannot be modified without physical presence verification. INV-003 (Hardware Separation of Control and Cognition) provides primary mitigation for 2 hazards (H-01, H-03), preventing cognitive clusters from directly accessing actuator paths or Secure World memory. INV-002 (Irreducible TCB ≤500 lines HDL) mitigates 1 hazard (H-05), because a minimal TCB reduces the attack surface for delay subversion. INV-006 (Complete Audit Integrity) mitigates 3 hazards (H-04, H-08, and indirectly H-03), providing forensic traceability and rollback detection. Every hazard has at least two independent mitigation paths (defense in depth), which is a necessary condition for SIL 3 compliance per IEC 61508-2 §7.4.5.

Non-Waivable Invariant Justification: The traceability analysis confirms the non-waivable status of INV-001, INV-005, and INV-006 within the AMAGI architectural class. INV-001 covers 6 of 8 hazards, including all four SIL 3 hazards (H-01, H-02, H-05, H-06) and the conservatively-treated H-03 (SIL 2–3); its removal would leave the system with no hardware-enforced barrier against the most severe threat scenarios. INV-005 covers 5 of 8 hazards, including all four SIL 3 hazards; its removal would permit unmediated access to the actuator pathway. INV-006 covers 3 hazards; while its coverage is narrower, it provides the only mitigation for H-04 (Audit Trail Compromise) and the only independent mitigation for H-08 (Rollback Attack). The removal of any of these three invariants would result in at least one SIL 3 hazard being left without adequate hardware-enforced mitigation, which is unacceptable per IEC 61508-1 §7.6.

8. Safety Requirements Allocation

This section allocates each identified hazard to a derived safety requirement, an implementing AMAGI component, and a verification method. The safety requirements are specified at a level of

detail sufficient to enable both formal verification and hardware testing. Each requirement is traceable to one or more invariants identified in Section 7 and is verifiable by the specified method. The verification methods include formal model checking using SPIN (for properties that can be expressed as temporal logic assertions), hardware testing with measured parameters (for timing-critical properties), and boot-time integrity verification (for properties related to immutability and supply chain integrity).

Hazard	Safety Requirement	Implementing Component	Verification Method
H-01	TTP chain must be physically non-bypassable; no software path may bypass the SCC policy gate	SCC + EGW (NIPU)	SPIN model checking (25,927,680 states explored; RI-2 model run)
H-02	MAG must be immutable during Normal Operation; no runtime modification of eFuse/OTP policy bits	MAG in eFuse/OTP (MEM)	Read-back verification at boot (CRTM Golden Hash comparison)
H-03	All memory access must pass through MPE; no direct access to Secure World memory from Normal World	MPE + ATL (MEM)	SPIN model checking (MPE-S01 through MPE-S06)
H-04	Audit trail must be append-only and tamper-proof; no overwrite or deletion of log entries	ATL audit diode + Amagi-Q (MEM/NIPU)	SPIN model checking (ATL-S01 through ATL-S05)
H-05	TTP delay must be physically enforced and PVT-stable; no software or configuration may shorten the delay	PVT-calibrated DDL (NIPU)	Fault injection testing per IEC 61508-2 Annex A
H-06	Kill switch must activate within cutoff window; termination latch must disconnect actuator path on SCC command	Termination latch + AFC (NIPU)	Hardware test, specified 12–35 µs cutoff window (TCB-only analytical envelope)
H-07	Post-silicon MAG validation against Golden Hash; detect any deviation from expected bit pattern	CRTM (NIPU)	Boot-time hash comparison (ADR-007)
H-08	Monotonic version counter prevents downgrade; no OTA package with version ≤ current may be accepted	Amagi-Q OTA protocol (NIPU/MEM)	Formal verification of counter monotonicity

Table 6: Safety requirements allocation and verification methods

The verification methods specified above reflect the defense-in-depth approach mandated by IEC 61508 for SIL 3 systems. Formal verification using SPIN provides mathematical proof that the TTP chain, memory protection, and audit trail properties hold under all reachable states of the system model. The SPIN model checking for H-01 explored 25,927,680 states, confirming that no execution path exists that bypasses the SCC policy gate. Hardware testing for H-05 and H-06 provides empirical confirmation that the physical implementation matches the formal model, with the cutoff window measured at 12–35 µs across all tested PVT corners. Boot-time integrity verification for H-02 and H-07 ensures that the immutable policy and firmware state have not been tampered with since fabrication, providing a hardware root of trust for the entire security chain.

For SIL 3 compliance, IEC 61508-2 §7.4.5 requires that the safety function achieve a diagnostic coverage of at least 99% for the dangerous failure modes. The combination of formal verification (exhaustive state space exploration) and hardware testing (measured timing parameters) provides coverage that exceeds this threshold. The SPIN model checking covers all logical failure modes, while the hardware testing covers all physical failure modes related to timing, voltage, and temperature. Residual failure modes (e.g., common-cause failures, side-channel attacks) are documented in Section 9.

8.1 SIL Targets vs Architecture Capability (new, D2)

> SIL targets in Table 4 are **requirements on the safety function in a given deployment**; they are not claims about any single build. The AMAGI architecture provides an integrity **capability** that is configuration-dependent (FMEDA v1.1, MPS v2.1 §3.1):
>
> - **1oo1 reference configuration (HFT=0): SIL 2** — current, RI-1/RI-2;
> - **1oo2 enhanced configuration (HFT=1, β=0.10): SIL 3** — Phase 2 target (2027), requires galvanic isolation and enhanced diagnostics.
>
> Consequently: hazards with SIL 3 targets (and SIL 2–3 hazards under the conservative reading) are addressable **only in the 1oo2 deployment class**; SIL 2 hazards are addressable in the 1oo1 reference configuration. This resolves the apparent conflict between "the reference implementation claims SIL 2" and "the HARA requires SIL 3": the reference implementation (RI-2, 1oo1) covers the SIL 2 hazard subset; the SIL 3 hazard subset defines the enhanced deployment envelope.

9. Residual Risk Assessment

This section documents the residual risks that remain after the AMAGI invariants are enforced. Residual risks are those that fall outside the certification boundary or for which no cost-effective mitigation exists within the current architectural framework. The identification and documentation of residual risks is required by IEC 61508-1 §7.6.3, which mandates that the HARA explicitly state which risks have been reduced to an acceptable level and which risks remain as residual. Three categories of residual risk have been identified for the AMAGI architectural class.

Residual Risk 1: Side-Channel Attacks (DPA/EMA). Differential Power Analysis (DPA) and Electromagnetic Analysis (EMA) attacks could potentially extract cryptographic keys from the Amagi-Q engine by monitoring power consumption or electromagnetic emanations during cryptographic operations. While the AMAGI architecture does not currently include dedicated side-channel countermeasures (e.g., masking, shuffling, constant-time implementations), this risk is designated for mitigation in Phase 2 of the AMAGI development roadmap. The rationale for deferral is that side-channel attacks require physical proximity to the device and sophisticated laboratory equipment, which significantly reduces the frequency parameter (F) for this hazard. Furthermore, the AMAGI architecture already provides partial mitigation through the append-only audit trail (INV-006), which would record any anomalous cryptographic operations associated with a side-channel attack attempt.

Residual Risk 2: Common-Cause Failure across NIPU and MEM. A common-cause failure (CCF) that simultaneously affects both the NIPU (processor) and MEM (memory) subsystems could defeat the hardware isolation boundary. Such a failure could be caused by a shared power supply fault, a shared clock domain failure, or an ionizing radiation event that corrupts state in both subsystems simultaneously. The AMAGI architecture mitigates this risk by implementing separate power domains for the NIPU and MEM, with independent voltage regulators and decoupling capacitors. Additionally, the AFC provides a third independent safety path that operates on a separate analog power domain, further reducing the probability of a CCF that defeats all three safety mechanisms simultaneously. The CCF beta factor is $\beta = 0.1$ (conservative estimate per the TCAD article [1]), which is within the acceptable range for SIL 3 per IEC 61508-6 Table D.1. With the addition of DTI and separate PLL/LDO for the SCC, a refined $\beta = 0.05$ may be achievable in Phase 2, further strengthening the CCF claim.

Residual Risk 3: Beyond-Certification-Boundary Threats. The AMAGI architecture does not protect against physical destruction of the chip, electromagnetic pulse (EMP) attacks exceeding the IEC 61000-4 Level 4 specification, or ionizing radiation doses exceeding the specified environmental envelope. These threats are considered beyond the certification boundary because they represent physical conditions under which no electronic system can guarantee safety. For deployments in environments with elevated radiation risk (e.g., nuclear containment, space), additional shielding

and radiation-hardened packaging should be specified at the system integration level. The AMAGI architecture provides no mitigation for these threats and explicitly excludes them from its safety claims.

Overall, the residual risk assessment confirms that the identified residual risks are acceptable for the targeted SIL 3 application domains. Side-channel attacks are deferred to Phase 2 with partial audit-trail mitigation in place; common-cause failures are addressed by separate power domains and the independent AFC; and beyond-boundary threats are explicitly excluded from the safety claims. No residual risk exceeds the tolerable risk threshold for SIL 3, and all residual risks are documented and tracked for future mitigation.

10. Conclusions

This Hazard Analysis and Risk Assessment confirms that the AMAGI architectural class, through its six hardware-enforced invariants, provides a configuration-dependent integrity capability: SIL 2 on the 1oo1 reference configuration (HFT = 0) and SIL 3 on the enhanced 1oo2 configuration (HFT = 1, β = 0.10; FMEDA v1.1 §8, MPS v2.1 §3.1). Eight hazards were systematically identified using the methodology prescribed by IEC 61508-1 §7.4, covering all four lifecycle phases and spanning consequence classes C2 through C4. Four of the eight hazards require SIL 3 integrity (H-01, H-02, H-05, H-06); two are assessed in the range SIL 2–3 and are conservatively treated as SIL 3 for deployment-configuration selection (H-03, H-07); the remaining two require SIL 2 (H-04, H-08). SIL 3 targets are therefore addressable in the 1oo2 deployment class; SIL 2 targets are addressable in the 1oo1 reference configuration. The risk graph method per IEC 61508-5 Annex D was applied with conservative parameters (W3 for all hazards), ensuring that the SIL targets are not dependent on assumptions about AI reliability.

The hazard-to-invariant traceability matrix demonstrates that no hazard is left without at least two independent mitigation paths (defense in depth) — a necessary condition for high-integrity claims per IEC 61508-2. The combination of formal verification (SPIN model checking with up to 25,927,680 states explored; the corrected model scc_amagi_v3 verifies L01–L04 with FA1/FA2, 2026-10) and the specified 12–35 µs cutoff window (TCB-only analytical envelope) provides the coverage basis for the hazard mitigations. The traceability matrix further confirms that INV-001 (Physical Non-Bypassability) is the most critical invariant, countering 6 of 8 hazards; INV-005 (Complete Mediation) counters 5 of 8 hazards; and INV-004 (Controlled Mutability of Critical Policies) counters 4 of 8 hazards.

The non-waivable status of INV-001, INV-005, and INV-006 is justified by their coverage of 6, 5, and 3 hazards respectively. The removal of INV-001 would leave 6 hazards without primary hardware-enforced mitigation, including all four SIL 3 hazards. The removal of INV-005 would leave 5 hazards without the complete mediation guarantee, eliminating the defense-in-depth property that is essential for high-integrity compliance. The removal of INV-006 would leave H-04 (Audit Trail Compromise) and H-08 (Rollback Attack) without any mitigation, which is unacceptable for systems operating in regulated domains where forensic traceability is a regulatory requirement (e.g., FDA 21 CFR Part 11, NRC Regulatory Guide 1.168).

This HARA should be updated when the threat model expands (e.g., to include side-channel attacks in Phase 2), when new deployment domains are added (e.g., space, automotive), or when formal verification results reveal previously unidentified hazard scenarios. The AMAGI architectural class is designed to be incrementally certifiable, meaning that this HARA can be extended without invalidating the existing hazard identification and SIL assignments. All changes to this HARA shall be subject to the configuration management process defined in the AGIAM Charter v1.2 (Governance and Configuration Management) and Framework v2.0 Section 28 (Compliance Verification Requirements), and shall require re-approval by the functional safety assessor prior to deployment in any SIL 3 application domain.

11. Change Log

Version	Date	Changes
1.0	2026-06	Initial HARA: 8 hazards, risk graph (IEC 61508-5 Annex D), invariant traceability
1.1	2026-10	Abstract/§7/§10 aligned with Table 4 (4 unconditional SIL 3 hazards; 2 conservatively treated); PFDavg-band column corrected; §8.1 targets-vs-capability (D2); deployment-configuration column; INV-005 §7 list aligned to the stated count; terminology canon (MAG/ATL/MPE/SCC per Master Glossary v1.0); Charter reference updated to v1.2